Q1/ Perform any 10 pandas operations or commands on the dataset after importing it in python.

```
!pip install pandas
```

```
Requirement already satisfied: pandas in .\anaconda3\Lib\site-packages
(3.0.3)
Requirement already satisfied: numpy>=2.3.3 in .\anaconda3\Lib\site-
packages (from pandas) (2.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in .\anaconda3\
Lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in .\anaconda3\Lib\site-packages
(from pandas) (2026.2)
Requirement already satisfied: six>=1.5 in .\anaconda3\Lib\site-
packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

10 OPEARTOPONS IN PANDAS

```
import pandas as pd
DataSheet = pd.read_csv(r"C:\Users\Ayesha Bro\Downloads\Data-Sheet-
f500-Physical - Data-Sheet-f500-Physical.csv")
#Opration No 1
print(DataSheet.shape)
#Opeartion No 2
print(DataSheet.head(5))
#opeartion No 3
print(DataSheet.tail(5))
#Operation No 4
print(DataSheet.describe())
#Operation No 5
print(DataSheet.loc[20])
#Operation No 6
#indexing and specific value access
DataSheet.iloc[10:20:2]
#operation No 7
pd.DataFrame(DataSheet)
print(type(DataSheet))
#Operation Np 8
display(DataSheet)
print(type(DataSheet["country"]))
display(DataSheet["country"])
#Opeartion No 9
display(DataSheet.info())
#Opearation No 10
DataSheet["sector"]
```

```
(500, 17)
                    Company  rank  revenues  revenue_change  profits
assets  \
0                    Google     1    485873             0.8  13643.0
```

```
198825
1                State Grid     2    315199            -4.4   9571.3
489838
2             Sinopec Group     3    267518            -9.1   1257.9
310726
3  China National Petroleum     4    262573           -12.3   1867.5
585619
4              Toyota Motor     5    254694             7.7  16899.3
437575

   profit_change                   ceo                    industry  \
0           -7.2  C. Douglas McMillon     General Merchandisers
1           -6.2              Kou Wei                Utilities
2          -65.0            Wang Yupu        Petroleum Refining
3          -73.7        Zhang Jianhua        Petroleum Refining
4          -12.3          Akio Toyoda  Motor Vehicles and Parts

                   sector  previous_rank country      hq_location  \
0              Retailing              1     USA  Bentonville, AR
1                 Energy              2   China   Beijing, China
2                 Energy              4   China   Beijing, China
3                 Energy              3   China   Beijing, China
4  Motor Vehicles & Parts              8   Japan    Toyota, Japan

                        website  years_on_global_500_list
employees  \
0        http://www.walmart.com                        23    2300000

1        http://www.sgcc.com.cn                        17     926067

2        http://www.sinopec.com                        19     713288

3        http://www.cnpc.com.cn                        17    1512048

4  http://www.toyota-global.com                        23     364445


   total_stockholder_equity
0                     77798
1                    209456
2                    106523
3                    301893
4                    157210
                          Company  rank  revenues  revenue_change
profits  \
495  Teva Pharmaceutical Industries   496     21903            11.5
329.0
496        New China Life Insurance   497     21796           -13.3
743.9
497       Wm. Morrison Supermarkets   498     21741           -11.3
```

```
406.4
498                              TUI   499     21655            -5.5
1151.7
499                        AutoNation   500     21609             3.6
430.5

     assets  profit_change                ceo  \
495   92890          -79.3   Yitzhak Peterburg
496  100609          -45.6            Wan Feng
497   11630           20.4      David T. Potts
498   16247          195.5   Friedrich Joussen
499   10060           -2.7  Michael J. Jackson

                            industry              sector
previous_rank  \
495                  Pharmaceuticals         Health Care
0
496  Insurance: Life, Health (stock)          Financials
427
497             Food and Drug Stores  Food & Drug Stores
437
498                  Travel Services   Business Services
467
499              Specialty Retailers           Retailing
0

     country           hq_location                       website  \
495   Israel  Petach Tikva, Israel     http://www.tevapharm.com
496    China        Beijing, China  http://www.newchinalife.com
497  Britain     Bradford, Britain     http://www.morrisons.com
498  Germany      Hanover, Germany      http://www.tuigroup.com
499      USA   Fort Lauderdale, FL    http://www.autonation.com

     years_on_global_500_list  employees  total_stockholder_equity
495                         1      56960                     33337
496                         2      54378                      8507
497                        13      77210                      5111
498                        23      66779                      3006
499                        12      26000                      2310
             rank       revenues  revenue_change       profits
assets  \
count  500.000000     500.000000      498.000000    499.000000
5.000000e+02
mean   250.500000   55416.358000        4.538353   3055.203206
2.436323e+05
std    144.481833   45725.478963       28.549067   5171.981071
4.851937e+05
min      1.000000   21609.000000      -67.300000 -13038.000000
3.717000e+03
25%    125.750000   29003.000000       -5.900000    556.950000
```

```
3.658850e+04
50%    250.500000   40236.000000         0.550000   1761.600000
7.326150e+04
75%    375.250000   63926.750000         6.975000   3954.000000
1.805640e+05
max    500.000000  485873.000000       442.300000  45687.000000
3.473238e+06

       profit_change  previous_rank  years_on_global_500_list
employees  \
count     437.000000     500.000000                 500.000000
5.000000e+02
mean       24.076888     222.134000                  15.036000
1.339983e+05
std       437.010425     146.941961                   7.932752
1.700878e+05
min      -793.700000       0.000000                   1.000000
3.280000e+02
25%       -22.700000      92.750000                   7.000000
4.293250e+04
50%        -0.400000     219.500000                  17.000000
9.291050e+04
75%        17.400000     347.250000                  23.000000
1.689172e+05
max      8909.500000     500.000000                  23.000000
2.300000e+06

       total_stockholder_equity
count                 500.000000
mean                30628.076000
std                 43642.576833
min                -59909.000000
25%                  7553.750000
50%                 15809.500000
75%                 37828.500000
max                301893.000000
Company                                      Ford Motor
rank                                                 21
revenues                                         151800
revenue_change                                      1.5
profits                                          4596.0
assets                                           237951
profit_change                                     -37.7
ceo                                     James P. Hackett
industry                          Motor Vehicles and Parts
sector                              Motor Vehicles & Parts
previous_rank                                        21
country                                             USA
hq_location                                Dearborn, MI
```

```
website                            http://www.ford.com
years_on_global_500_list                            23
employees                                       201000
total_stockholder_equity                         29170
Name: 20, dtype: object
<class 'pandas.DataFrame'>

                              Company  rank  revenues  revenue_change
profits  \
0                              Google     1    485873             0.8
13643.0
1                          State Grid     2    315199            -4.4
9571.3
2                      Sinopec Group      3    267518            -9.1
1257.9
3          China National Petroleum     4    262573           -12.3
1867.5
4                       Toyota Motor     5    254694             7.7
16899.3
..                                ...   ...       ...             ...
...
495  Teva Pharmaceutical Industries   496     21903            11.5
329.0
496        New China Life Insurance   497     21796           -13.3
743.9
497       Wm. Morrison Supermarkets   498     21741           -11.3
406.4
498                              TUI   499     21655            -5.5
1151.7
499                       AutoNation   500     21609             3.6
430.5

     assets  profit_change                  ceo  \
0    198825           -7.2  C. Douglas McMillon
1    489838           -6.2              Kou Wei
2    310726          -65.0            Wang Yupu
3    585619          -73.7        Zhang Jianhua
4    437575          -12.3          Akio Toyoda
..      ...            ...                  ...
495   92890          -79.3    Yitzhak Peterburg
496  100609          -45.6             Wan Feng
497   11630           20.4       David T. Potts
498   16247          195.5    Friedrich Joussen
499   10060           -2.7   Michael J. Jackson

                             industry                 sector
previous_rank  \
0              General Merchandisers               Retailing
1
1                          Utilities                  Energy
```

```
2
2                  Petroleum Refining                   Energy
4
3                  Petroleum Refining                   Energy
3
4           Motor Vehicles and Parts  Motor Vehicles & Parts
8
..                                ...                      ...
...
495                   Pharmaceuticals              Health Care
0
496  Insurance: Life, Health (stock)               Financials
427
497              Food and Drug Stores      Food & Drug Stores
437
498                   Travel Services       Business Services
467
499               Specialty Retailers               Retailing
0

     country           hq_location                         website  \
0        USA       Bentonville, AR        http://www.walmart.com
1      China        Beijing, China        http://www.sgcc.com.cn
2      China        Beijing, China        http://www.sinopec.com
3      China        Beijing, China        http://www.cnpc.com.cn
4      Japan         Toyota, Japan  http://www.toyota-global.com
..       ...                   ...                          ...
495   Israel  Petach Tikva, Israel      http://www.tevapharm.com
496    China        Beijing, China   http://www.newchinalife.com
497  Britain     Bradford, Britain      http://www.morrisons.com
498  Germany      Hanover, Germany       http://www.tuigroup.com
499      USA   Fort Lauderdale, FL     http://www.autonation.com

     years_on_global_500_list  employees  total_stockholder_equity
0                          23    2300000                     77798
1                          17     926067                    209456
2                          19     713288                    106523
3                          17    1512048                    301893
4                          23     364445                    157210
..                        ...        ...                       ...
495                         1      56960                     33337
496                         2      54378                      8507
497                        13      77210                      5111
498                        23      66779                      3006
499                        12      26000                      2310

[500 rows x 17 columns]

<class 'pandas.Series'>
```

```
0          USA
1        China
2        China
3        China
4        Japan
        ...
495     Israel
496      China
497    Britain
498    Germany
499        USA
Name: country, Length: 500, dtype: str

<class 'pandas.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 17 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Company                  500 non-null    str
 1   rank                     500 non-null    int64
 2   revenues                 500 non-null    int64
 3   revenue_change           498 non-null    float64
 4   profits                  499 non-null    float64
 5   assets                   500 non-null    int64
 6   profit_change            437 non-null    float64
 7   ceo                      500 non-null    str
 8   industry                 500 non-null    str
 9   sector                   500 non-null    str
 10  previous_rank            500 non-null    int64
 11  country                  500 non-null    str
 12  hq_location              500 non-null    str
 13  website                  500 non-null    str
 14  years_on_global_500_list  500 non-null    int64
 15  employees                500 non-null    int64
 16  total_stockholder_equity  500 non-null    int64
dtypes: float64(3), int64(7), str(7)
memory usage: 119.2 KB

None

0                  Retailing
1                     Energy
2                     Energy
3                     Energy
4      Motor Vehicles & Parts
               ...
495              Health Care
496               Financials
497        Food & Drug Stores
498         Business Services
```

```
499                 Retailing
Name: sector, Length: 500, dtype: str
```

Q2/A teacher wants to develop a Python quiz system that randomly selects questions for students and records their results with the date and time.

```
import random
Questions =  {
    "What is the capital of Afghanistan?": "kabul",
    "How many days are in a month?": "30",
    "Which language am i  learning?": "python",
    "What is 5 + 6?": "11",
    "Which planet do we live on?": "mars"
}
#Writing a function start_quiz(questions) that:
def Start_Quiz(Questions):
    selected_Questions = random.sample(list(Questions.keys()), 3)
    score = 0

    for Question in selected_Questions:
        print(Questions)
        answer= input("Enter your answer: ")

        if answer.strip().lower() == Questions[Question].lower():
            print("Correct!")
            score += 1
        else:
            print("Correct Answer:", Questions[Question])
            print("Incorrect!")

    print("Your final score is:", score, "out of 3")
    return score
Start_Quiz(Questions)
```

```
{'What is the capital of Afghanistan?': 'kabul', 'How many days are in
a month?': '30', 'Which language am i  learning?': 'python', 'What is
5 + 6?': '11', 'Which planet do we live on?': 'mars'}

Enter your answer:  kabul

Correct Answer: python
Incorrect!
{'What is the capital of Afghanistan?': 'kabul', 'How many days are in
a month?': '30', 'Which language am i  learning?': 'python', 'What is
5 + 6?': '11', 'Which planet do we live on?': 'mars'}

Enter your answer:  11

Correct!
{'What is the capital of Afghanistan?': 'kabul', 'How many days are in
```

```
a month?': '30', 'Which language am i  learning?': 'python', 'What is
5 + 6?': '11', 'Which planet do we live on?': 'mars'}

Enter your answer:  python

Correct Answer: 30
Incorrect!
Your final score is: 1 out of 3

1
```

Part B — Generate a Quiz Report (Continuation of Q2)

```python
from datetime import datetime

def generate_report(name, score):
    percentage = (score / 3) * 100

    if percentage == 100:
        result = "Excellent"
    elif percentage == 67 or percentage == 33:
        result = "Good"
    elif percentage == 66.67 or percentage == 33.33:
        result = "Good"
    else:
        result = "Needs Improvement"

    current_time = datetime.now()

    Report = {
        "Student": name,
        "Score": str(score) + "/3",
        "Percentage": f"{percentage:.2f}%",
        "Result": result,
        "Date and Time": current_time
    }

    print("\n----- QUIZ REPORT -----")
    for key, value in Report.items():
        print(key + ":", value)

generate_report("Ayesha",2)
```

```
----- QUIZ REPORT -----
Student: Ayesha
Score: 2/3
Percentage: 66.67%
Result: Needs Improvement
Date and Time: 2026-10-18 12:44:46.862162
```

Q3/a) You are required to build a simple student grade management system using Python. Store student names and their marks in a dictionary, where each student's name is a key and their marks in three subjects are stored in a list.

```
students = {
    "Zainab": [80, 75, 90],
    "Ayesha": [85, 88, 92],
    "Abeeha": [60, 70, 65],
    "Raza": [95, 90, 98]
}

print(students)
```

```
{'Zainab': [80, 75, 90], 'Ayesha': [85, 88, 92], 'Abeeha': [60, 70, 65], 'Raza': [95, 90, 98]}
```

b) calculate Student Results

```
def calculate_results(students):
    results = {}

    for name, marks in students.items():
        total = 0

        for mark in marks:
            total += mark

        average = total / len(marks)

        if average >= 90:
            grade = "A"
        elif average >= 80:
            grade = "B"
        elif average >= 70:
            grade = "C"
        elif average >= 60:
            grade = "D"
        else:
            grade = "Fail"

        results[name] = {
            "Average": average,
            "Grade": grade
        }

    return results
```

c) Find Top Performers

```python
def find_topper_students(students):
    top_students = []

    for name, marks in students.items():
        total = 0

        for mark in marks:
            total += mark

        average = total / len(marks)

        if average >= 80:
            top_students.append(name)

    return top_students


print("\nStudent Results:")
print(calculate_results(students))

print("\nTop Students:")
print(find_topper_students(students))
```

```
Student Results:
{'Zainab': {'Average': 81.66666666666667, 'Grade': 'B'}, 'Ayesha':
{'Average': 88.33333333333333, 'Grade': 'B'}, 'Abeeha': {'Average':
65.0, 'Grade': 'D'}, 'Raza': {'Average': 94.33333333333333, 'Grade':
'A'}}

Top Students:
['Zainab', 'Ayesha', 'Raza']
```

Q4/Design a Python program for a Library Book Management System using classes, objects, and inheritance. Create the following classes:

A. Class Book (4 Marks)

```python
class Book:
    def __init__(self, title, author, price,genre):
        self.title = title
        self.author = author
        self.price = price
        self.genre = genre

    def display_info(self):
        print("Book Title:", self.title)
        print("Author:", self.author)
        print("Price:", self.price)
        print("genre:", self.genre)
```

B. Class EBook (4 Marks)

```
class EBook(Book):
    def __init__(self, title, author, price, file_size, genre):
        super().__init__(title, author, price, genre)
        self.file_size = file_size

    def display_info(self):
        super().display_info()
        print("File Size:", self.file_size, "MB")
```

C. Objects and Output (4 Marks)

```
book1 = Book("Whimsical", "Zainab Raza",2000,"nature")
ebook1 = EBook("Data Science", "Jamal khan", 8000,5,"Learning")

print("--- BOOK DETAILS ---")
book1.display_info()

print("--- EBOOK DETAILS ---")
ebook1.display_info()
```

```
--- BOOK DETAILS ---
Book Title: Whimsical
Author: Zainab Raza
Price: 2000
genre: nature
--- EBOOK DETAILS ---
Book Title: Data Science
Author: Jamal khan
Price: 8000
genre: Learning
File Size: 5 MB
```